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5. The table gives information about some radioactive isotopes.

Radioactive isotope	Radiation emitted	Half-life	Use
sodium-24	beta and gamma	15 hours	detecting leaks in underground pipes
technetium-99m	gamma	6 hours	radioactive tracer which is injected into blood
iridium-192	gamma	74 days	detecting cracks inside steel plates
francium-223	alpha or beta	22 minutes	no uses
palladium-103	beta	17 days	treating tumours by direct injection
radon-222	alpha	3.8 days	early warning for earthquakes

Use the information to answer the questions below.

(a) The time for radioactive isotopes to decay to a safe level is different for each one.

(i) State which radioactive isotope decays most quickly. .... [1]

(ii) **Complete the table below**, which shows the time to decay to a fraction of $\frac{1}{32}$. [3]

Fraction remaining	Time		
	Palladium-103	Sodium-24	Technetium-99m
1	0	0	0
$\frac{1}{2}$	17 days	15 hours	....
$\frac{1}{4}$	34 days	30 hours	....
$\frac{1}{8}$	51 days	45 hours	....
$\frac{1}{16}$	68 days	....	....
$\frac{1}{32}$	85 days	75 hours	....

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(b) (i) Explain why iridium-192 is suitable for detecting cracks inside steel plates. [2]

....

....

....

(ii) Explain why palladium-103 is suitable for treating cancer by directly injecting it into a tumour. [2]

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(iii) I. Radon-222 gas can be dangerous when breathed in.
Explain why. [2]

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II. State why radon-222 levels vary across the UK. [1]

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